

CP2+ Explained

How a safe genomic-feasibility check helps an exact path solver avoid impossible searches

directed possibility + genomic possibility =
useful search

Project

Bioinformatics Exact Methods Demo Cockpit

Generated

June 25, 2026

Scope

Educational implementation and deterministic evidence

Audience

Students, supervisors, jury members, and researchers

EXECUTIVE SUMMARY

CP2+ in under one minute

CP2+ in one sentence.

CP2+ keeps CP2's exact longest-path search and adds a safe early test that stops a branch when no still-reachable genomic bridge can reconnect the selected reactions.

Problem

Find the longest valid simple directed path in acyclic D whose selected reactions induce one connected neighborhood in genomic graph G .

Addition

Before extending a branch, test whether selected genomic components can still reconnect through unused vertices reachable from the current endpoint.

Preserved

Same objective, directed bound, deterministic lexical winner, validity rule, exactness semantics, and proof-complete requirement.

Evidence

16 deterministic cases; 0 equality mismatches; 35 safe genomic prunes; 417 CP2 states versus 232 CP2+ states.

Non-claim

Structural evidence only: reduction in 3 cases, equality in 13, increase in 0. No universal runtime claim.

What the evidence supports. On this fixed educational corpus, completed CP2+ runs preserved the exact result while sometimes ending demonstrably hopeless branches earlier. It supports the integration and its traceable behavior—not a victory parade over every solver on every machine.

READER MAP

From metabolic city to sound pruning

CP2 is a detective following one-way roads in a metabolic city. CP2+ is the same detective, carrying one extra map: before walking deeper into a branch, it checks whether the selected genomic neighborhoods could still be connected by any vertex that remains reachable ahead.

The short version. CP2+ solves the same exact longest (D,G) -consistent simple-path problem as CP2. It adds a safe necessary-condition filter. It does not change the objective, deterministic tie-break, proof semantics, or meaning of a valid answer.

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 directed arc in D  undirected link in G  selected path  forward reachable  impossible branch

1. TWO GRAPHS, ONE SET OF REACTIONS

The biological-network story

Imagine a city whose locations are biochemical reactions. The same reactions appear on two maps because each map answers a different question.

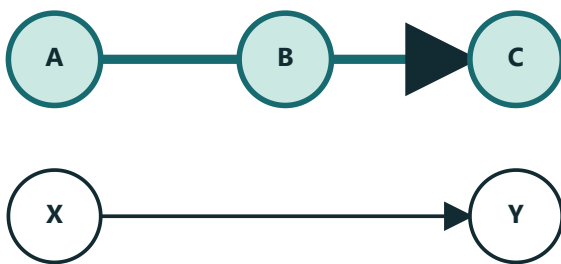
Metabolic graph D

D is directed and acyclic. An arc $u \rightarrow v$ means reaction v can follow reaction u in the modeled metabolic ordering. Direction matters. A road behind the detective cannot be used as a right extension.

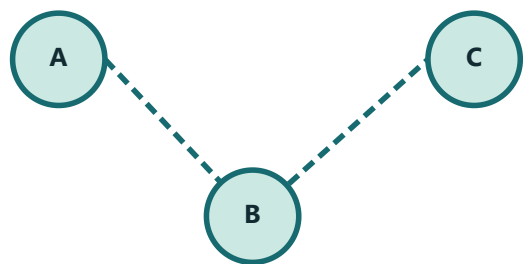
Genomic graph G

G is undirected. A link $u - v$ represents genomic proximity between the genes associated with reactions u and v . It says "nearby," not "before" or "after."

D: METABOLIC ORDER



G: GENOMIC PROXIMITY



The vertex identity is shared: reaction A is A on both maps. The edges encode different relations.

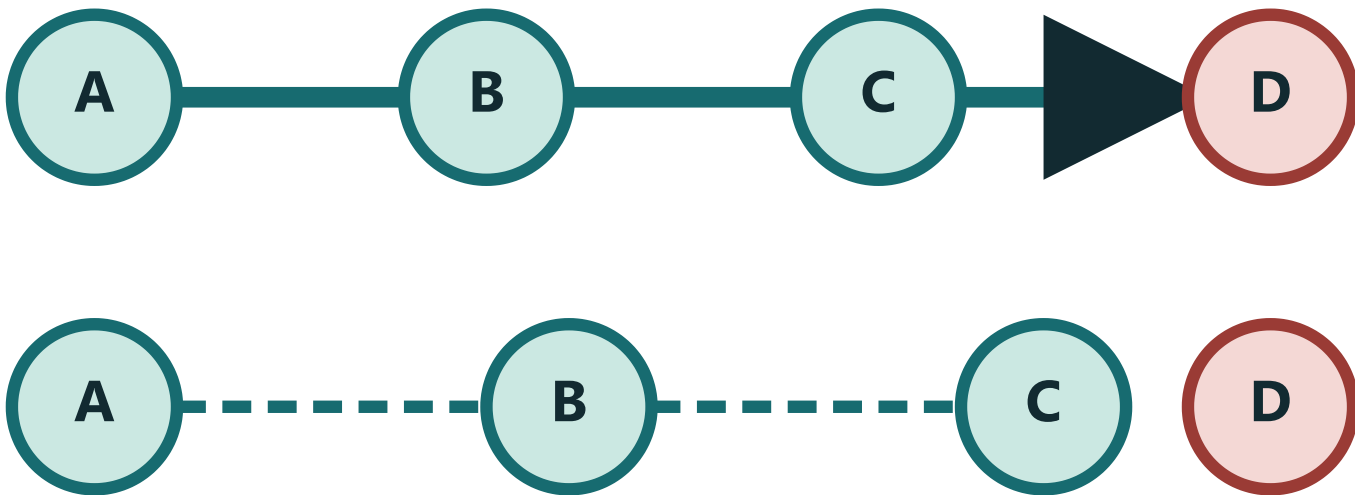
Why share vertices? Because the final object must satisfy both views at once. We choose reactions that form a legal one-way metabolic path in D, and those same chosen reactions must form one connected genomic neighborhood in the subgraph induced by them in G.

2. ONE OBJECTIVE, TWO CONSTRAINTS

The core challenge

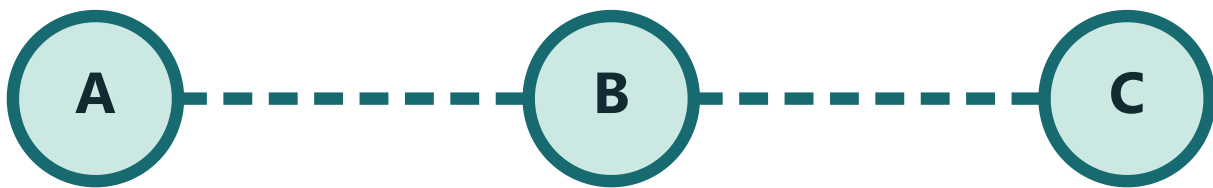
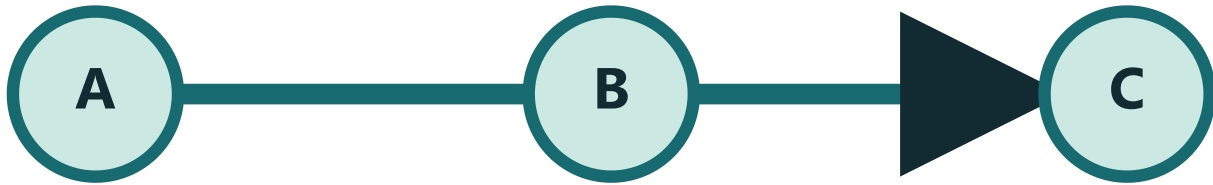
A solution is a **simple directed path** $P=(v_1, \dots, v_k)$ in D : every consecutive pair is a directed arc and no vertex repeats. The selected vertex set $V(P)$ must induce a **connected** subgraph $G[V(P)]$. The objective is to maximize $|P|$, the number of selected vertices. Equal-length optima use the existing deterministic lexicographic winner.

Long in D , invalid in G



$A \rightarrow B \rightarrow C \rightarrow D$ has length 4, but D is isolated in the induced genomic subgraph. It is not a valid answer.

Shorter, valid, and therefore better



A→B→C has length 3 and its induced genomic subgraph is connected. It beats every invalid longer candidate.

Detective rule: the longest road itinerary is irrelevant if its selected neighborhoods do not form one connected genomic district.

3. FAMILY PORTRAIT

CP1, CP2, CP2+, CP3, CP4 — do not mix them up

Method	Problem setting	What it adds	Correct mental model
CP1	Simple paths in acyclic directed D	Basic constraint-programming path model and genomic validation	The detective explores legal directed paths.
CP2	Same problem as CP1	A safe directed longest-suffix upper bound	Stop when the remaining road is too short to beat the incumbent.
CP2+	Same exact problem and winner as CP2	Safe forward genomic-feasibility propagation	Also stop when no reachable future neighborhood can reconnect the selection.
CP3	Cyclic directed graphs; maximum-coverage trails	Eulerian-subgraph style modeling for trails	A different trip type: edges may form cyclic trail structures.
CP4	Same cyclic trail setting as CP3	Walk-and-cover branching	CP3 with a search strategy favoring uncovered vertices.

Boundary worth defending: CP2+ is an implementation-level integration of CP2’s directed longest-path propagation with safe genomic-feasibility filtering. It does not turn CP2 into CP3 or CP4, is not presented as a new research method, and is not a paper reproduction.

DETAILED DIFFERENCE GUIDE

Exactly What CP2+ Offers Beyond CP2 and Other Methods

A. CP2 versus CP2+

Topic	CP2	CP2+
Problem solved	Longest valid simple path in acyclic D	Same
Objective	Maximize path length	Same
Tie-break	Deterministic lexical winner	Same
Directed bound	Yes	Yes, unchanged
Early genomic impossibility check	No	Yes
New pruning question	"Can the road ahead still be long enough?"	"Can any usable future vertex still reconnect the genomic neighborhoods?"
Exactness on completed search	Yes	Yes
Universal speed claim	No	No

The bridge example

Suppose the current path has selected reactions in two genomic neighborhoods. The directed graph still offers several vertices ahead, so CP2 says, "There may still be enough road ahead." CP2+ replies, "Fine, but can the neighborhoods ever become one district?" If every possible genomic bridge is behind the current endpoint or otherwise unreachable, the answer is no and CP2+ safely stops.

If a reachable bridge remains, CP2+ keeps searching. It is an impossibility detector, not an optimism detector wearing a lab coat.

- CP2+ does **not** search a different objective.
- CP2+ does **not** alter the winning path rules.
- CP2+ does **not** automatically make every instance faster.
- CP2+ adds a necessary-condition prune only.

RELATED GENOMIC REASONING, DIFFERENT MACHINERY

CP2+ versus AlgoBB++, CP1, ILP, and Subset

B. CP2+ versus AlgoBB++

AlgoBB++ is a separate branch-and-bound architecture with its own relaxed genomic reconnection logic. CP2+ does not copy AlgoBB++ as a whole. It applies a one-ended, forward-reachability version of related feasibility thinking inside the CP2 path-search workflow. Think of two detectives consulting related neighborhood maps while investigating with different procedures; neither has stolen the other’s trench coat.

Question	CP2+	AlgoBB++
Search architecture	CP2-style one-ended path search	Branch-and-bound / arc-by-arc style
Genomic view	Selected vertices + forward-reachable unused vertices	Its own relaxed reconnection logic
Goal in this project	Educational CP2 enhancement	Contrasting exact baseline
Claim	Integration-level enhancement	Existing exact method

C. CP2+ versus CP1

- CP1 = explore
- CP2 = explore less when the road is too short
- CP2+ = also stop when the future neighborhoods cannot reconnect

CP1 is the basic CP educational model. CP2 adds directed longest-suffix pruning. CP2+ keeps that bound and adds safe genomic-feasibility pruning. The progression changes how early hopeless work is recognized, not what counts as the winning answer.

D. CP2+ versus ILP1, ILP2, and Exact Subset DP

These methods aim at the same acyclic-path destination using different vehicles:

ILP1

A path formulation with connectivity witness variables. It expresses selection and genomic connectivity through an educational integer-programming-style model.

ILP2

A rooted connectivity witness using parent links and levels. Same destination, a more explicit family tree for genomic connection.

Exact Subset DP

State-space dynamic programming over selected subsets and endpoints. It remembers visited combinations instead of pruning a CP2 search tree.

CP2+ is a pruning-enhanced search approach, not a replacement for these alternative exact educational formulations. Same destination, different vehicle; nobody needs to flatten the tires.

DO NOT MERGE THE PROBLEM TRACKS

CP2+ versus CP3 and CP4

Acyclic path track:
CP1 → CP2 → CP2+

Cyclic trail track:
CP3 → CP4

CP2+ solves simple paths in acyclic D. CP3 and CP4 solve maximum-coverage trails in cyclic D. A trail may revisit vertices but cannot repeat arcs.

CP3 and CP4 are therefore not “better versions” of CP2+. They answer a different class of question. Calling CP4 an upgrade to CP2+ would be like praising a ferry for being a newer bicycle: both move things, but the route rules have changed.

CP2+ Does Not Magically...

- solve cyclic trail problems; CP3 and CP4 cover that family;
- prove universal speed superiority;
- replace all exact methods;
- guarantee that every genomically connected relaxed graph has a valid directed completion;
- claim to be a new research method;
- reproduce a published benchmark or implementation; or
- make capped or cancelled runs exact—CP2+ is exact only when search completes.

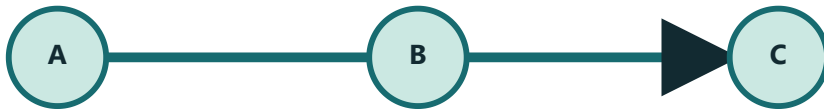
The useful claim is narrower: when the optimistic forward genomic graph cannot reconnect the selected vertices, no legal right extension can repair them. That is enough to prune safely, and no cape is required.

4. THE ORIGINAL PRUNING ENGINE

CP2 in plain language

At a partial path, CP2 computes a safe upper bound from the best possible directed suffix still available. If the current path plus even the most optimistic road ahead cannot improve the incumbent—or cannot win the existing lexical tie rule—the branch can be discarded.

INCUMBENT: LENGTH 3



NEW BRANCH: AT MOST LENGTH 2



road ahead too short

The detective does not need to inspect every lamp post on D→X. Even perfect luck cannot produce a path longer than the incumbent.

This is **DIRECTED BOUND PRUNE**. It concerns directed length, not genomic repair. CP2+ keeps this mechanism unchanged and applies the genomic check only where the directed bound has not already settled the branch.

5. THE ADDITIONAL SAFE QUESTION

The CP2+ idea

Let P be the current simple directed path, last its endpoint, and U the unused vertices. CP2+ collects the unused vertices reachable by a directed path starting at last :

$$F(P, \text{last}) = \{u \in U : u \text{ is forward-reachable from last in } D \text{ without reusing } P\}$$

It then builds a deliberately relaxed genomic graph containing the selected vertices and every possible forward-reachable helper:

$$H(P, \text{last}) = G[V(P) \cup F(P, \text{last})]$$

This relaxation is generous. It allows all reachable future helpers at once, even if no single directed completion could necessarily use all of them together. That generosity is the source of safety: failing an optimistic test proves impossibility; passing it merely says “do not prune yet.”

Safe genomic-feasibility pruning theorem

Assumptions. D is directed and acyclic; extensions append unused vertices to the right of P ; genomic validity requires $G[V(P')]$ connected for a completed extension P' .

Soundness statement. If the vertices of P are not in one connected component of $H(P, \text{last})$, no right extension of P can become genomically connected.

Proof sketch. Every vertex that any legal right extension can append is forward-reachable from last , so it belongs to $F(P, \text{last})$. Therefore every completed extension induces a genomic subgraph contained in $H(P, \text{last})$. If even this larger, optimistic graph cannot connect the already selected vertices, no smaller realizable extension can connect them. Pruning cannot remove a feasible completion.

Important asymmetry. Connected in H means “possibly repairable,” not “guaranteed completable.” Disconnected in H means “provably impossible.” CP2+ uses only the second implication.

Eight detailed examples

Each card uses the released teaching-library concept. Solid arrows are metabolic direction; dashed links are genomic proximity. Green vertices are selected, gold vertices are reachable future helpers, and red marks an impossible continuation.

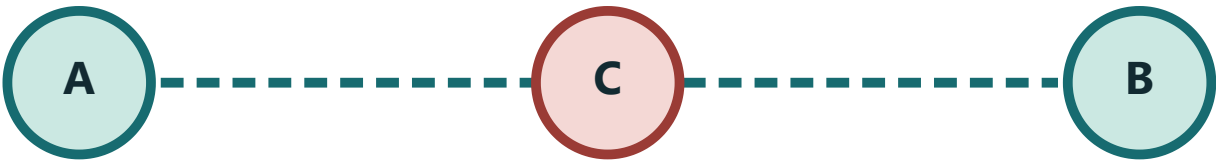
1. Unreachable genomic bridge

SAFE GENOMIC PRUNE

D



G

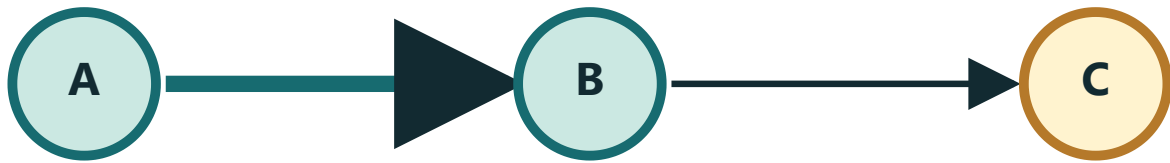


Path	A→B
Forward set	empty
CP2 sees	A legal directed prefix.
CP2+ adds	The only genomic bridge C cannot be reached after B.
Why?	A bridge painted on a map is useless when no forward road reaches it.

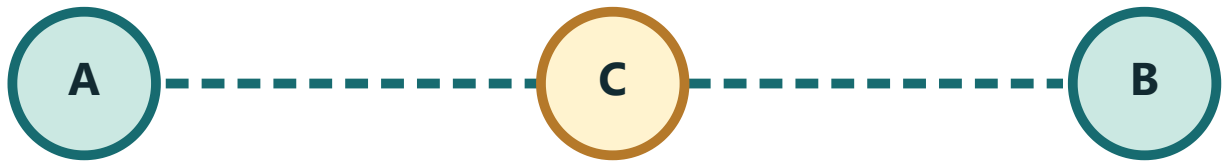
2. Forward-reachable genomic bridge

NO PRUNE

D



G



Path A→B

Forward set {C}

CP2 sees A legal prefix not yet genomically valid.

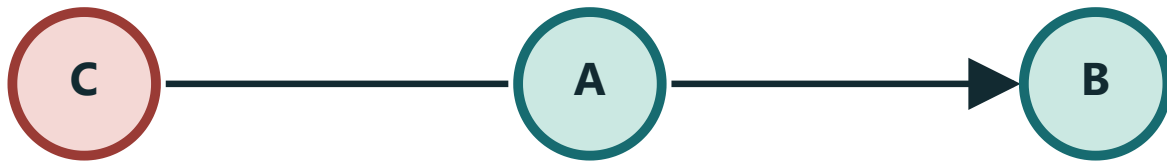
CP2+ adds C reconnects A and B in the relaxed graph.

Why? Disconnected now does not mean disconnected forever. The detective keeps walking.

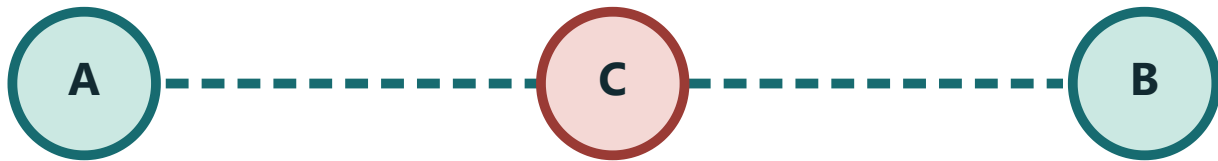
3. Backward-only genomic bridge

SAFE GENOMIC PRUNE

D



G



Path	A→B
Forward set	empty
CP2 sees	A legal one-ended prefix.
CP2+ adds	C exists, but only before A; it cannot be appended after B.
Why?	A bridge behind you is not useful when the road only goes forward.

4. Already-connected prefix

NO PRUNE

D



G

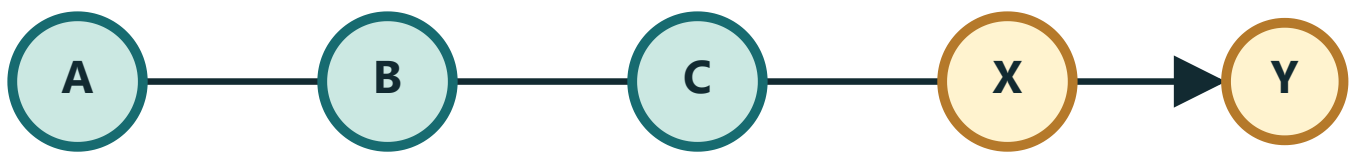


Path	A→B→C
Forward set	{D}
CP2 sees	A length-improvable branch.
CP2+ adds	The selected prefix is already connected.
Why?	The extra check is allowed to say "all clear." Safe propagation is not compulsory pruning.

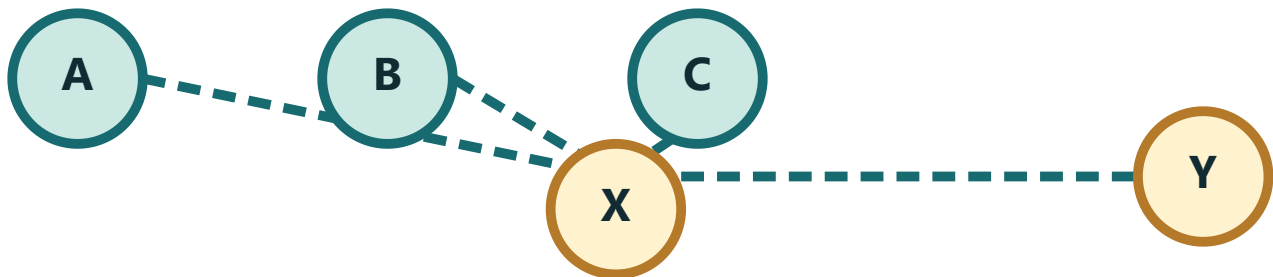
5. Multi-component recovery

NO PRUNE

D



G



Path A→B→C

Forward set {X,Y}

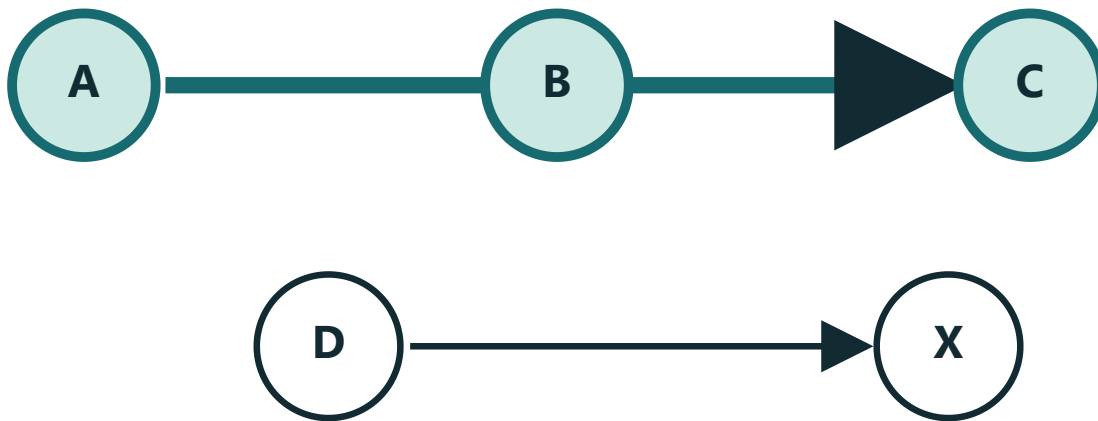
CP2 sees Three selected vertices with directed room ahead.

CP2+ adds X can join all three genomic components.

Why? The relaxed graph keeps branches that are complicated but still repairable.

6. Directed dead end

DIRECTED BOUND PRUNE

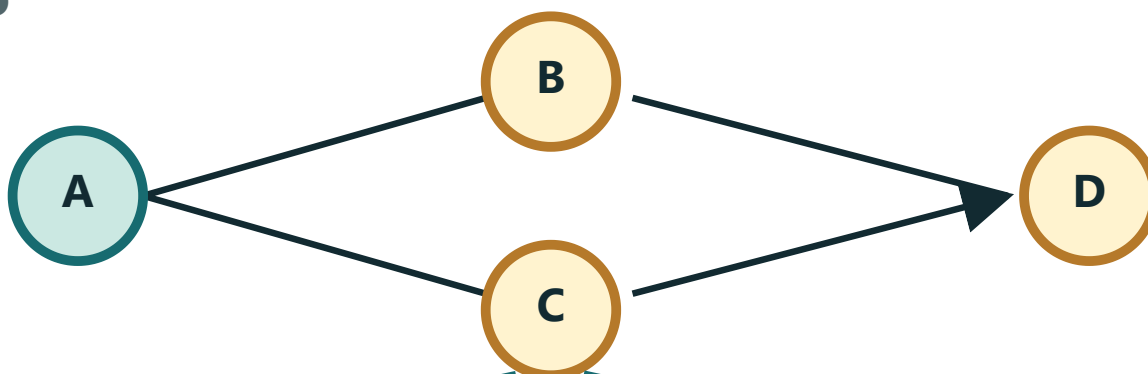


Path	D
Forward set	{X}
CP2 sees	Maximum possible length 2 versus incumbent length 3.
CP2+ adds	Nothing is needed; CP2's bound acts first.
Why?	Do not perform a neighborhood survey after the road-length inspector has closed the branch.

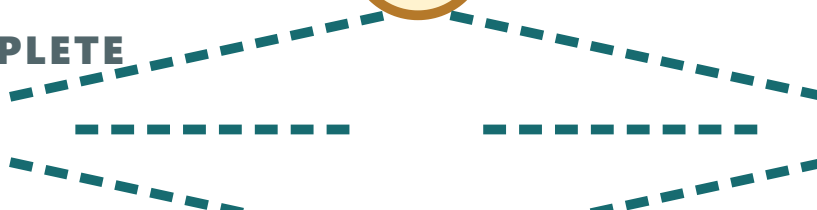
7. Dense genomic graph

NO PRUNE

D



G COMPLETE

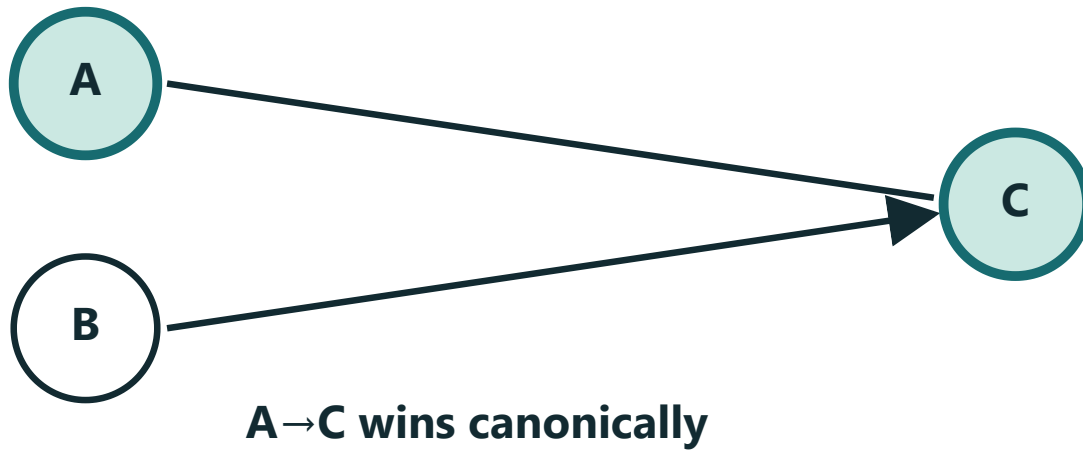


Path	various prefixes
Forward set	branch-dependent
CP2 sees	Directed alternatives and bounds.
CP2+ adds	Checks usually pass because G is dense.
Why?	More doors between neighborhoods mean fewer impossibility certificates—and the checks still cost work.

8. Lexicographic tie

TIE-BREAK

D



Candidates A→C and B→C

Objective Both length 2

CP2 sees Equal optimal lengths; apply deterministic lexical rule.

CP2+ adds No change to ordering or winner.

Why? A pruning improvement may shorten the investigation, but it may not rewrite the judge's rulebook.

The two most important non-obvious cases

A. Repairable disconnected prefix

The selected vertices may currently be disconnected in $G[V(P)]$. That alone is not a pruning reason. Future selected vertices can connect existing components because induced connectivity is checked on the final selected set.

For $A \rightarrow B \rightarrow C$ with genomic links $A-C-B$, the prefix $A \rightarrow B$ is disconnected, yet C is forward-reachable and repairs it. CP2+ sees A and B connected inside H , so it correctly returns **NO PRUNE**.

B. Backward-only bridge

A bridge can exist in the full genomic graph and still be unusable. With $C \rightarrow A \rightarrow B$, C lies behind endpoint B . A one-ended right extension cannot travel backward and insert C before A .

Therefore C is absent from $F(P, \text{last})$. If A and B remain disconnected without it, pruning is safe. This is why “reachable somewhere in D ” is too weak; the test requires reachable *forward from the current endpoint*.

Memory aid: CP2+ asks neither “is the prefix connected now?” nor “does some bridge exist anywhere?” It asks “can any vertex still usable from here reconnect the selected components?”

Comparison with other exact methods

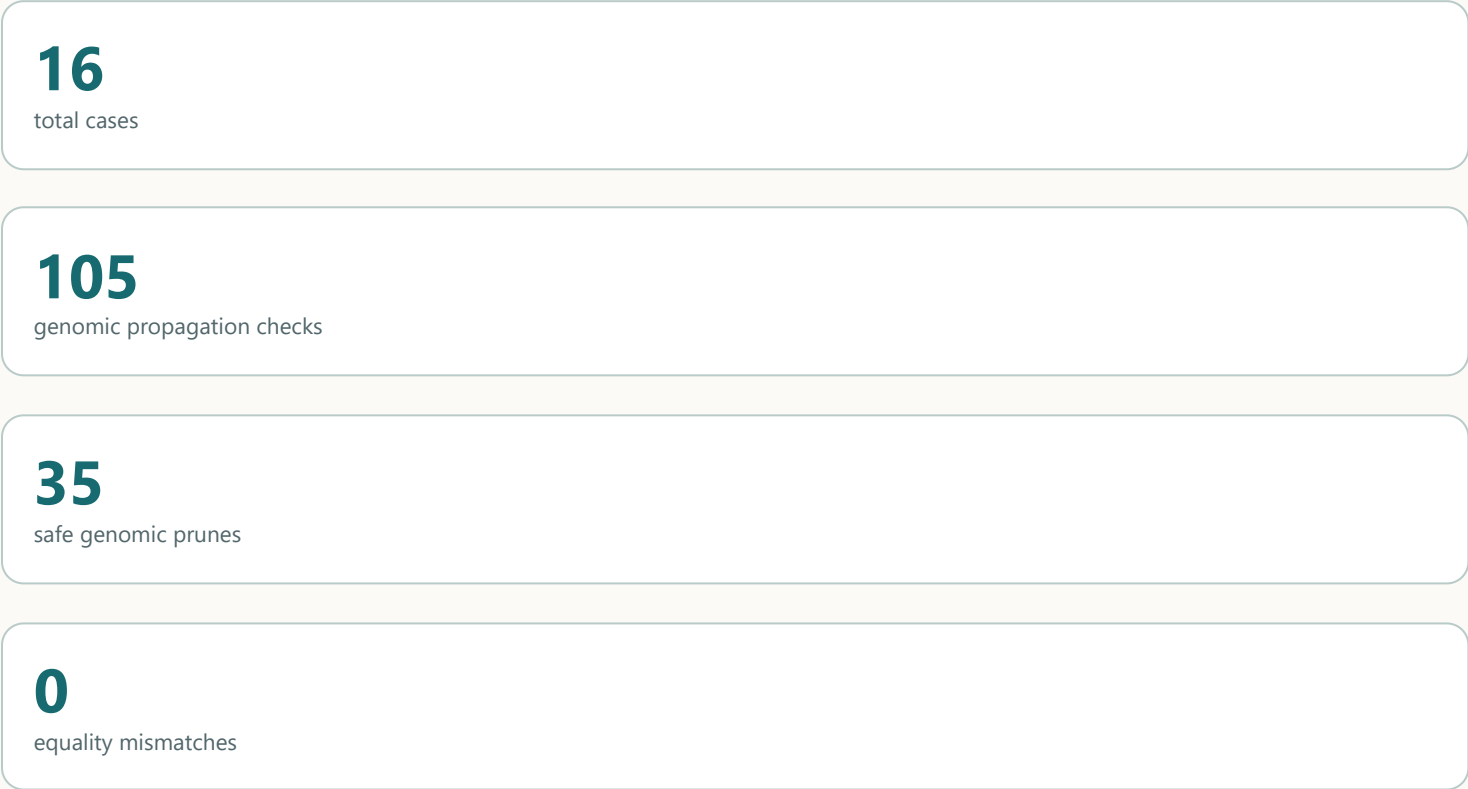
Method	Problem / graph	Path or trail	Main technique	Genomic connectivity	Educational role	Limitations
Legacy	Acyclic D	Simple path	Exhaustive enumeration	Validate induced G on candidates	Transparent exact baseline	Combinatorial growth
CP1	Acyclic D	Simple path	Constraint-style path search	Candidate feasibility	Basic CP formulation	Small educational instances
CP2	Acyclic D	Simple path	CP1 plus directed upper bound	Candidate feasibility	Shows safe directed pruning	No early genomic impossibility filter
CP2+	Acyclic D	Simple path	CP2 plus one-ended safe genomic feasibility	Relaxed graph on selected and forward-reachable vertices	Demonstrates integrated safe propagation	Check overhead; exact only on completion
AlgoBB++	Small directed graphs in project scope	Path search	Branch-and-bound, arc-by-arc growth	Its own relaxed two-ended genomic reconnection logic	Contrasting exact search architecture	Not the same integration as CP2+
ILP1	Small DAGs	Simple path	Educational binary x/y/z formulation	Connectivity witness variables	Shows an ILP-style exact model	No claim of native MIP or CPLEX execution
ILP2	Small DAGs	Simple path	Root, parent links, levels	Rooted genomic connectivity witness	Alternative educational formulation	No claim of OR-Tools or production MIP
Exact Subset DP	Small DAGs	Simple path	Dynamic programming on subset and endpoint	Exact state/candidate checks	Educational exact implementation	Exponential subset space; not a novel method
CP3	Cyclic directed graphs	Trail	Maximum coverage / Eulerian-subgraph model	Trail-oriented genomic constraints	Reference concept for cyclic problems	Different problem, not a direct CP2+ alternative
CP4	Cyclic directed graphs	Trail	CP3 plus walk-and-cover branching	As in cyclic trail model	Search-strategy reference	Different problem, not a CP2+ replacement

This project does not claim CPLEX, OR-Tools, native MIP execution, genome-scale production capability, a new research method, or reproduction of a published implementation.

9. RELEASED DETERMINISTIC EVIDENCE

Evidence Benchmark

The benchmark contains 16 deterministic educational and bounded-stress cases across fragmented, dense, repairable bridge, small exhaustive, and larger bounded-stress families. Its primary evidence is structural. Wall-clock timing is intentionally secondary because browser and test-machine timing is unstable.



States explored



Structural reduction: 185 explored states. This is not labeled a runtime speedup.

Events emitted



Structural reduction: 1005 trace events. The bars visualize released counts, not fabricated percentages.

Evidence item	Released result	Interpretation
Reduced / equal / increased state cases	3 / 13 / 0	Reduction is concentrated, not universal.
Objective mismatches	0	Completed benchmark objectives agree.
Winner mismatches	0	Deterministic winning paths agree.
Proof-status mismatches	0	Proof-complete status agrees.
Validity mismatches	0	Path and induced genomic validity agree.

Generated benchmark conclusion: Across 16 deterministic educational and bounded-stress cases, CP2+ reduced states in 3, matched CP2 state counts in 13, and explored more states in 0. It emitted 35 safe genomic propagation prunes. Exactness comparison recorded 0 equality mismatches. These structural results are corpus-specific; runtime was not used as a scientific conclusion.

Honest interpretation

Where CP2+ can help

Fragmented genomic cases are the natural target. Directed search may offer many continuations, while the genomic map proves that some selected components can never reconnect using anything reachable ahead. Those branches can end early.

Where it may be neutral

When G is dense, selected sets are often connected or easily repairable. CP2+ performs checks but obtains few genomic prunes. CP2's directed bound may already do most useful pruning.

Why overhead is real

Forward reachability, component inspection, and trace events require work. Fewer search states do not automatically imply lower wall-clock time on every machine or fixture.

Why exactness comes first

A faster wrong answer is not progress. The important property is that the new filter is sound and completed runs preserve objective, canonical winner, validity, and proof status.

The released evidence is intentionally modest: a small deterministic corpus can demonstrate behavior, regression safety, and structural differences. It cannot establish genome-scale performance, broad real-world distributions, or universal runtime superiority.

Traceability and proof behavior

Structural counters

States explored	Search states actually visited.
Directed-bound prunes	Branches closed by CP2's length/lexical bound.
Genomic checks	Relaxed forward-feasibility inspections.
Genomic prunes	Checks that proved reconnection impossible.

Trace journal

Each event records the evolving path, bound, forward frontier, genomic components, counters, and reason. The journal and current-event card use the same canonical event index.

STEP current / total

This ordinal stays synchronized with Previous, Next, Play, End, Reset, example changes, graph state, and the active trace row.

Proof-complete means search completed

An exact optimal/no-solution claim requires completed search and a proof-complete event. A cap or cancellation returns an **incomplete** result. An incumbent may be useful, but it is not proof of optimality. CP2+ is exact only when search completes.

Never promote an interrupted run. "Best found so far" and "proven optimal" are different scientific statements.

Final recap — what CP2+ adds

- **Same exact objective:** longest simple directed path in acyclic D whose selected vertices induce a connected subgraph in undirected G.
- **Same deterministic winner:** CP2+ preserves the existing lexical tie-break.
- **Extra safe early rejection:** prune only when all forward-reachable unused vertices still cannot reconnect selected genomic components.
- **Evidence-backed teaching:** deterministic structural benchmark, trace journal, explicit counters, and cross-method equality checks.
- **No universal promise:** dense cases may be neutral, checks have overhead, and corpus-specific reductions are not universal runtime superiority.

One-sentence takeaway: CP2+ is CP2 with an additional optimistic genomic map; when even that generous map cannot connect the selected neighborhoods, the detective can safely stop investigating the branch.

Formal definitions

- Input.** A finite shared vertex set V ; an acyclic directed graph $D=(V,A)$; and an undirected genomic-proximity graph $G=(V,E)$.
- Feasible path.** A sequence $P=(v_1,...,v_k)$ of distinct vertices such that $(v_i,v_{i+1})\in A$ for every consecutive pair and $G[\{v_1,...,v_k\}]$ is connected.
- Objective.** Maximize $k=|P|$; break equal-length ties using the project’s deterministic lexicographic comparison.
- Forward frontier.** $F(P,last)$ contains unused vertices reachable from endpoint $last$ by a directed continuation that does not reuse selected vertices.
- Relaxed feasibility graph.** $H(P,last)=G[V(P)\cup F(P,last)]$.
- Pruning rule.** If selected vertices lie in more than one connected component of $H(P,last)$, reject the branch. Otherwise retain it; passing the test is not a proof that a directed completion exists.

Notation glossary

Symbol	Meaning
D	Acyclic directed metabolic-order graph
G	Undirected genomic-proximity graph
P	Current selected simple directed path
last	Current endpoint of P
U	Unused vertices
$F(P,last)$	Forward-reachable unused vertices
$H(P,last)$	Relaxed induced genomic graph on selected plus reachable future vertices
Incumbent	Best feasible path found so far; not necessarily proven optimal
Proof-complete	Search completed with exact status established

Benchmark protocol and limits

Protocol summary

The corpus is fixed in source control: no random generation, external data, wall-clock threshold, or machine-dependent input. Primary metrics are states explored, events emitted, directed-bound prunes, genomic propagation checks and prunes, exact-result equality, and proof-complete equality. Small exhaustive cases are checked against the available exact educational solvers; larger bounded cases compare CP2 and CP2+ structural behavior under existing cap semantics.

Allowed conclusions

- State or event counts were lower, equal, or higher on a stated corpus case.
- A family produced safe genomic prunes.
- Completed runs agreed or differed on objective, winner, validity, or proof status.
- Dense cases showed neutrality or check overhead in this corpus.

Limits and non-claims

- CP2+ is not universally faster or universally better.
- The fixtures do not establish broad real-world or genome-scale production performance.
- The work is not claimed as a new research method or paper reproduction.
- The project does not claim CPLEX, OR-Tools, native MIP execution, or production-scale optimization infrastructure.
- CP2+ remains a path solver for acyclic D; CP3/CP4 address cyclic trail settings.

Permanent limitation note: These results are specific to this deterministic educational and bounded-stress corpus. They do not establish universal runtime superiority or reproduce a published benchmark.

LIVE DEMO ACCESS

Try the Interactive Demo

The interactive page lets readers replay the trace, inspect safe genomic prunes, and compare structural benchmark evidence.

Public CP2+ route



[bioinformatics-exact-methods-demo.vercel.app/
methods/cp2-plus](https://bioinformatics-exact-methods-demo.vercel.app/methods/cp2-plus)

Public Comparison Lab



[bioinformatics-exact-methods-demo.vercel.app/
methods/cp2-plus/comparison](https://bioinformatics-exact-methods-demo.vercel.app/methods/cp2-plus/comparison)

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